

Alfred M. Mbaru

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PROFESSIONAL SUMMARY

Junior Data Scientist skilled in **Python** and **SQL** with hands-on experience building **supervised learning** and **time-series** models, and operationalizing **ETL/ELT** pipelines on **AWS (S3, Glue, Kinesis, Lambda, RDS)**. Proficient with **scikit-learn**, **pandas/NumPy**, and **Tableau/Power BI** for model development and stakeholder reporting. Demonstrated ability to translate business questions into features, models, and measurable KPIs.

TECHNICAL SKILLS

- **Languages/DB:** Python; **SQL (PostgreSQL)**; Bash
- **Machine Learning:** scikit-learn; XGBoost; Logistic/Random Forest; **Classification/Regression; Time Series (ARIMA, SARIMA, Prophet)**; Feature Engineering; **Cross-Validation; Hyperparameter Tuning (GridSearchCV)**; Model Evaluation (**ROC-AUC, PR-AUC, Precision, Recall, F1, MAPE/MAE**)
- **Data Engineering / MLOps:** **AWS (S3, Glue, Kinesis, Lambda, RDS)**; ETL/ELT; Batch + Streaming; API Integration; Web Scraping; Version Control (**Git/GitHub**); **Docker (basic)**
- **Analytics & BI:** Tableau; Power BI; Excel (Pivot Tables, Lookups); KPI Design; Dashboarding; Data Storytelling

AREA OF EXPERTISE

Data Analytics | Machine Learning | Predictive Modeling | Data Engineering | Cloud Computing (AWS) | Data Visualization | Data Storytelling | KPI Analysis | End-to-End Data Pipelines

PROJECTS & HACKATHONS

Customer Churn Prediction — Supervised Learning | Feb 2025

Built preprocessing (encoding, imputation, scaling) and trained Random Forest baseline; performed GridSearchCV tuning and cross-validation; communicated drivers to business stakeholders via dashboards.

AWS Recommender (Batch + Streaming) | Jul 2025

Orchestrated Glue jobs and Kinesis streams; transformed and materialized features in PostgreSQL using pgvector for similarity search; documented runbooks and failure recovery.

WiDS Datathon — fMRI Classification | Mar 2025

Engineered statistical features from neuroimaging data; compared classifiers; tracked experiments for reproducibility; reported model performance and limitations.

FinGuard Lite — Forecasting & BI | 2025 (Undergraduate Project)

Built forecasting components and trend dashboards for financial KPIs; automated refresh and alerting; delivered scenario analysis in Power BI/Tableau.

EDUCATION

BSc, Computer Science | Expected 2025 Kabarak University – Nakuru, Kenya

PROFESSIONAL CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services, 2025

Stanford Medicine: Fundamentals of Data Science in Precision Medicine and Cloud Computing, 2025

Unilever Supply Chain Data Analyst Professional Certificate – Coursera, 2025

RELEVANT TRAININGS

Professional Foundation-ALX Africa,2025

Data analytics-ALX Africa,2025

VOLUNTEER EXPERIENCE

Mentor, Data Science & ML Club – Kabarak University | 2023 – Present

- Guided and mentored students on foundational and advanced topics in Python, data analytics, and machine learning through hands-on projects.